EXHIBIT A.3.3.3.1: BASIC TOWER & SITE STRUCTURAL DESIGN PARAMETERS

5G FWA TRIAL DEMO (BAND n50) — NTC REGULATORY COMPLIANCE EXHIBIT

Operator / Applicant:	Telecommunications Technology Solutions Inc. (TelTech)	**Authorized Representative:**	OLIVER TUNGOL
Network Partner:	Comclark Network and Technology Corp.	**Structural / Site Engineer:**	JOMAR DOGILLO
System / Frequency:	5G SA FWA — Band n50 (1457.0 – 1517.0 MHz)	**Governing Codes:**	NSCP 2015 (7th Ed.), TIA-222-G/H, AISC 360-16

1. PHILIPPINE BASIC WIND SPEED & WIND ZONE MAPPING (NSCP 2015)

Per **National Structural Code of the Philippines (NSCP 2015 Section 207)** and PAGASA meteorological records, structural calculations and design wind forces are established according to Philippine Wind Zones:

Wind Parameter / Regime	Wind Zone 1 (Typhoon Track)	Wind Zone 2 (NCR & Central Luzon)	Wind Zone 3 (Low Risk Inland)
Geographic Coverage	Batanes, Cagayan, Isabela, Aurora, Bicol, Samar, Leyte, Surigao	**NCR, Pampanga, Bulacan, Tarlac, Bataan, CALABARZON, Cebu**	Palawan, Zamboanga, Davao, Maguindanao, Sulu
Ultimate Basic Wind Speed (V) (3-sec gust, 50-yr return, NSCP 2015)	**315 km/h (87.5 m/s)**	**270 km/h (75.0 m/s)**	**200 km/h (55.6 m/s)**
Nominal Basic Wind Speed (Vasd)	245 km/h (68.0 m/s)	210 km/h (58.3 m/s)	155 km/h (43.1 m/s)
Structure / Occupancy Category	Category IV (Standard Comms) / Category III (Essential)	Category IV / Category III	Category IV / Category III
Importance Factor (Iw)	1.15 (Essential) / 1.00 (Standard)	1.15 (Essential) / 1.00 (Standard)	1.15 (Essential) / 1.00 (Standard)
Terrain Exposure Category	Exposure C (Open) / D (Coastal Shore)	Exposure B (Suburban) / C (Open)	Exposure B / Exposure C

2. OPERATIONAL VS. SURVIVABLE WIND SPEED & DEFLECTION LIMITS

Engineering Parameter	Standard Specification / Threshold	Technical notes & Functional Compliance
Survivable Wind Speed (Vsurvival)	**270 km/h to 315 km/h** (3-second gust, Ultimate Limit State)	Tower, slip-joints, baseplate, and foundation anchorage withstand extreme typhoon wind forces under full equipment loading without plastic collapse.
Operational Wind Speed (Voperational)	**140 km/h to 160 km/h** (40.0 m/s to 44.4 m/s, Serviceability State)	Structure maintains rigid alignment during high-wind storms, ensuring continuous 5G n50 beamforming and backhaul RF link transmission.
Allowable Twist (θtwist)	≤ **±0.5° (30 arcmin)** for MW / Backhaul ≤ **±1.0° (60 arcmin)** for Macro Sectors	Maximum torsional rotation about vertical tower axis under operational wind pressure (140-160 km/h) per TIA-222-H.
Allowable Sway / Tilt (θtilt)	≤ **±1.0° (60 arcmin)** at top antenna mounting centerline	Angular deflection of vertical mast at 30.0m AGL. Prevents beam pattern decoupling over residential clusters.

Allowable Verticality (Plumbness)	≤ **1 : 500 (0.20% of Height H)** (Max **60 mm** for 30m tower)	Out-of-plumb deviation measured throughout full mast height in still air under dead load. Maximum slip-joint tolerance: 1:1000 (0.10%).
Top Horizontal Deflection (Δtop)	≤ **H / 100** (Max 300 mm for 30m tower)	Maximum lateral service deflection at mast tip under serviceability wind load.

3. TOWER, COMPOUND & FOUNDATION CIVIL SPECIFICATIONS

Tower Structure	30-Meter Heavy-Duty Polygonal Tubular Steel Monopole (12/18-sided tapered polygon, 3-4 slip-joint sections, min. overlap 1.5x mating diameter). Steel plate ASTM A572 Grade 50 (Fy = 345 MPa).
Corrosion Protection	Hot-dip galvanized per ASTM A123 / ISO 1461, coating thickness ≥ 85 µm (610 g/m²). Hardware per ASTM A153.
Foundation & Concrete	Reinforced Concrete Pad & Chimney (Mat Foundation) or Cast-in-place Bored Piles. 28-day fc ≥ 28 MPa (4,000 psi). Deformed rebar ASTM A615 Grade 60 (Fy = 414 MPa). Safety Factor ≥ 2.0 bearing, ≥ 1.5 overturning.
Anchor Bolts	ASTM F1554 Grade 55 or Grade 105 heavy-duty anchor bolts with dual heavy-hex leveling and lock nuts.
Compound Layout	8.0m x 8.0m (64 m²) gravel-topped compound, 2.4m cyclone fence with 3 strands razor wire, steel entrance gate.
Earthing & Lightning	Franklin lightning rod extending ≥ 2.0m above top antenna, 70 mm² copper down-conductor, perimeter ground ring with 4-6 copper-clad ground rods (19mm x 3.0m). Earth resistance ≤ 5.0 Ohms.